

Raga Mouni Batchu

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SUMMARY

Experienced software/AI Engineer with 6+ years in designing, developing, and maintaining scalable solutions. Skilled in cloud technologies, backend development, with research experience applying GraphRAG and agentic AI techniques. Proven expertise in migrating legacy systems to microservices, enhancing performance, and leading high-impact teams. Served as Scrum Master for 3 years, delivering consistent results through agile practices.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, C++, Perl

AI/ML Frameworks & Libraries: PyTorch, Hugging Face Transformers, LangChain, Vertex AI, PySpark

AI/ML Techniques: Supervised Learning, Instruction-Based Fine-Tuning, Retrieval-Augmented Generation (RAG), GraphRAG, AI Agent Development, LLM Evaluation, Prompt Engineering

Databases: PostgreSQL, MongoDB, DynamoDB, SQL

Backend & APIs: Spring Boot, Node.js, GraphQL, Hasura, RESTful API, FastAPI, OAuth 2.0, SAML

Cloud Platforms: AWS (S3, SQS, RDS, SSM, EC2, ECS, CloudFront, CloudWatch, Lambda, EventBridge, Bedrock)

Infrastructure & DevOps: Docker, Kubernetes, AWS CDK, Terraform, Terragrunt, Jenkins, CI/CD, Git

Data & Messaging: Apache Kafka, Informatica, RabbitMQ

Monitoring, Logging & API Tools: Grafana, Prometheus, Tableau, Graylog, Postman, cURL

WORK EXPERIENCE

PennText Education, Pennsylvania, United States of America

Lead Software/AI Engineer

05/2026 – Current

- Owned the full lifecycle of a multi-modal LLM extraction pipeline for LAYERS, PennText's AI platform: framed the problem, evaluated candidate models on output quality versus latency and per-conversation cost, and shipped a 3-tier fallback cascade that cut cost ~ 68% within a 150s SLA.
- Built CoursePackPlus, an AI course-authoring platform, implementing a multi-turn LLM interview pipeline and a RAG enrichment layer that pulls in research to generate structured, publishable coursepacks, with GitHub-webhook sync into the LAYERS data store.
- Delivered CRM and lead-pipeline automation, including an automated lead-notification pipeline (serverless edge functions + Gmail API) that unifies contacts, interactions, and sales-pipeline data for the sales team.

iPipeline, Pennsylvania, United States of America

AI Engineer Intern

01/2026 – 05/2026

- Architected an agentic digitization service leveraging AWS Bedrock and Python to map complex paper-based forms to digital templates via high-dimensional vector similarity matching.
- Engineered a scalable Event-Driven Architecture using Terraform, EventBridge, and SQS to orchestrate asynchronous document processing, ensuring 100% data durability and enabling automated metadata storage in DynamoDB.
- Built high-performance MCP servers to ingest and validate insurance data, transforming unstructured inputs into structured representations for downstream automation.
- Integrated LLM tool-calling capabilities with MCP servers to automate form-filling and quote retrieval, creating a seamless bridge between natural language instructions and legacy insurance APIs

athenahealth, Bengaluru, India

Senior Software Engineer (Consumer Messaging Zone)

06/2023 – 08/2024

- Led the migration of call-related features from a legacy system to microservices, adding new functionalities and saving \$732k annually by reducing manual client team workloads.
- Designed and implemented a comprehensive monitoring system for messaging services, optimizing the operational visibility of the microservices.
- Created dashboards in Graylog on-demand basis, which eased the debugging of production issues
- Improved SMS opt-in rates for patients by 30% by integrating a DTMF node in call flows.
- Spearheaded a team of 5 engineers, resolving critical production issues and ensuring consistent system uptime.
- Served as Scrum Master for 3 years, achieving excellent sprint performance and delivery records.

- Resolved long-standing system issues, including memory leaks and critical 5xx errors due to ulimit.

Software Engineer (Consumer Messaging Zone) 06/2021 – 05/2023

- Developed an email blacklist feature that eliminated hard bounces, significantly improving KPR.
- Extended SMS services to support Spanish, increasing accessibility for the US-based user base.
- Designed and developed a vendor driver library, enabling % based traffic migration with multi-vendor partitioning.
- Built a Two-Way SMS service, facilitating 100,000 patient replies per day for appointment confirmations, cancellations, or rescheduling.

Associate Software Engineer (Consumer Messaging Zone) 06/2019 – 05/2021

- Implemented a last-minute reminder system for patient appointments, reducing no-show rates by 18%.
- Resolved high-priority bugs impacting production systems, improving product stability and client satisfaction.

TiVo, Bengaluru, India 06/2018 – 07/2019

Software Engineer Intern

- Enhanced the TV on-screen guide search experience, improving usability and content discoverability.
- Developed features to preserve user state during navigation, ensuring a seamless transition when returning to previous screens.

EDUCATION

West Chester University of Pennsylvania 01/2025 - 05/2026

Master of Science in Computer Science (GPA: 4/4)

Shri Vishnu Engineering College for Women 06/2015 - 05/2019

Bachelor of Technology in Computer Science (GPA: 9/10)

RESEARCH & PUBLICATIONS

Bidirectional Translation of ASL and English using CNN and Transformers **Published**

- Built a bidirectional ASL-English translation system combining a CNN-BiLSTM sign-to-gloss model, Transformer-based gloss-to-sentence models, and a gloss-to-sign animation pipeline.

A Survey on Data Visualization Techniques for International Trade **Published**

ASL recognition and game-based interaction for Deaf Learners **Published**

- Developed an ASL learning game with a custom backtracking-based puzzle generator and a spatial-temporal Transformer for gloss recognition, achieving 89% Top 5 accuracy on WL-ASL dataset.

KG-RAG for Tox21: Explainable Drug Toxicity Prediction **Submitted**

- Developing a knowledge graph-based RAG application on the toxicity domain, integrating Tox21, PubChem, DSSTox, ChemBL, UniProt and Reactome data products.

PROJECTS

- Novel Generator using LLM Agentic Framework and RAG Pipeline:** Designed a tool that takes user-provided information, generates novel chapters using an LLM Agentic framework and a RAG pipeline, extracts metadata from the generated content, and stores it in a vector database for use in subsequent chapter generations.
- Fine-Tuning Flan-T5 for Entity Extraction:** Fine-tuned the Flan-T5 model on a custom instruction-based entity extraction dataset using LoRA, improving extraction accuracy by approximately 25% compared to the original Flan-T5 model. [GitHub](#)
- Organ Donor & Recipient Matching DApp:** Developed a decentralized application using Ethereum Solidity and Web 3.0 to match organ donors with recipients.

CERTIFICATIONS & ACHIEVEMENTS

- AWS AI Practitioner Certificate
- Certification in “Generative AI with Large Language Models” – Coursera
- Outstanding Graduate Student Award by West Chester University of Pennsylvania CSM
- Awarded Starlight Award for 4 releases consecutively at athenahealth
- Winner, ConsenSys India Blockchain Hackathon 2018
- University Innovation Fellow, Stanford University